Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/02/2026
PI Name:	Robinson, Stephen K. Ph.D.		
Project Title:	HCAAM VNSCOR: Enabling Autonomous Crew Task Performance with Multimodal Electronic Procedure Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stephen.k.robinson@ucdavis.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	530-754-9495
Organization Name:	University of California, Davis		
PI Address 1:	Department of Mechanical & Aerospace Engineering		
PI Address 2:	1 Shields Avenue, 2132 Bainer Hall		
PI Web Page:			
City:	Davis	**State:**	CA
Zip Code:	95616-5270	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/15/2019	**End Date:**	06/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2025 per A. Beitman/JSC (Ed., 8/23/23) NOTE: End date changed to 12/23/2023 per V. Lehman/JSC and NSSC information (Ed., 7/20/23) NOTE: End date changed to 12/31/2023 per A. Beitman/JSC (Ed., 4/17/23) NOTE: End date changed to 4/14/2023 per NSSC information (Ed., 1/26/21) NOTE: End date changed to 3/14/2020 per NSSC information (Ed., 1/22/2020)		
Key Personnel Changes/Previous PI:	February 2021 report: Jessica Marquez, Ph.D., is now CoInvestigator (CoI). Steven Hillenius and Richard Joyce are no longer CoIs on the project.		

COI Name (Institution):	Karasinski, John M.S. (NASA Ames Research Center) Marquez, Jessica Ph.D. (NASA Ames Research Center)
Grant/Contract No.:	80NSSC19K0657
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR). Future long duration exploration missions (LDEM) conducted by NASA will have an increased need for crew autonomy during routine and emergency procedures, due to the increased distance from Earth causing time delays in communications. Presently, many procedures are completed with constant communication between the crewmembers and mission control personnel. This need for increased autonomy will lead to a need for more information being stored on board and accessed by crewmembers in a timely and context appropriate manner during procedural execution. Emergent technologies in multimodal interaction such as augmented reality (AR) visual displays, spatial audio, and tactile feedback are likely to play a role in mitigating this need, leading to what we define as "enhanced electronic procedures." In this proposal we outline a research study which will use a multimodal enhanced electronic procedure to determine the best tasks and cues to pair with sensory channels for procedural execution tasks. Past efforts by our group have investigated procedural tasks using new technologies such as augmented reality and haptic cues. A ground-based research study will determine the effects of crew performance, situational awareness, and trust with the use of multimodal enhanced electronic procedures compared to traditional unimodal electronic procedures. The results of the ground-based study will lead to deployment in an analog mission for validation in a flight-like environment. From the lab and analog results, we will formulate recommendations for updated standards and guidelines for multimodal interaction and electronic procedures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research aims to re-define the meaning of "procedures" for astronauts performing complex task in space. Traditionally, procedures are static, non-responsive documents that serve as passive instructions or recipes for astronauts to follow while performing a pre-planned task. The current research adds sensors and responsive procedure-viewer technology to allow a dynamic feedback loop to aid the astronaut in being certain that the procedures are being executed correctly. Since procedures are very common in safety-critical tasks here on Earth (operating rooms, nuclear power plants, airliners, etc.), the results of this NASA research are likely to benefit a broad range of society on Earth.
Task Progress:	This NASA-sponsored study explored how enhanced multimodal procedures impact complex mechanical repair task performance in an isolated spacecraft analog environment. Procedures were enhanced through embedded multi-modal guidance (instructional text, animations, photos, and videos with user-selected detail); these enhancements were integrated into a new software-based Electronic Procedure Viewer (EPV). While the EPV did not improve average workload or efficiency, the procedure did have the important benefit of significantly improving participants' task accuracy. This is a crucial finding; as space exploration missions travel farther from Earth, communication latency will restrict Mission Control Center (MCC) ability to monitor astronauts. Any tool that improves task accuracy significantly increases both the autonomy and safety of the onboard crew during daily operations. Furthermore, during the participants' first time completing the task, the Enhanced Procedure significantly improved both efficiency and accuracy, indicating that enhanced procedures are a powerful tool to help workers complete new or unfamiliar tasks more effectively. These data support the use of enhanced procedures for just-in-time training and off-nominal scenarios where both efficiency and accuracy are vital for mission success. An additional study in the project assessed the value of a tablet-based augmented reality (AR) procedure enhancement. Early results suggest that for complex and unfamiliar manual tasks, AR-based procedures may require less up-front task training, which is especially important for astronauts executing complex, long-duration missions in space. Key Points: ● Compared to traditional methods, Enhanced Procedures significantly and consistently improve task accuracy during complex mechanical repair tasks. ● Enhanced Procedures dramatically improve task performance (efficiency and accuracy) when applied to novel or unfamiliar tasks. ● To sustain user trust in procedural aids, provided sensor feedback must be robust and consistently accurate. ● Augmented reality-based procedures that are visually presented on a hands-free tablet (instead of on a head-mounted display) can guide astronauts through complex procedures with little previous specific task training.
Bibliography Type:	Description: (Last Updated: 06/02/2026)
Articles in Other Journals or Periodicals	Ben-Ora K, Homer K, Karasinski J, Marquez J, Robinson SK. "Multimodal procedural enhancements improve repair accuracy in a spacecraft analog." Human Factors. Submitted as of April 2026. , Apr-2026